ST. CLARET COLLEGE (AUTONOMOUS), BENGALURU

Department of Bachelor of Computer Applications (BCA)

Vacation Mini Project Report

Nimbus

Weather Forecaster

Track Your Skies, Anywhere.

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Semester	4th Semester
Submission	July 2026

https://weather-nimbus-nikita.netlify.app/

BCA Department · 4th Semester · July 2026

CERTIFICATE

This is to certify that the mini project entitled

NIMBUS — WEATHER FORECASTER

has been successfully developed and submitted by NIKITA (Roll No. 24BCA096), a student of the 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru.

This mini project was developed independently as part of self-learning during the vacation period. The work is original and has not been submitted elsewhere for any academic examination or award.

St. Claret College (Autonomous), Bengaluru

DECLARATION

I, NIKITA, Roll No. 24BCA096, 4th Semester, BCA, St. Claret College (Autonomous), Bengaluru, declare that the mini project “NIMBUS — WEATHER FORECASTER” is a record of original work done by me during the vacation period.

I declare that:

- This project was carried out independently, without any formal institutional guidance.
- The work is genuine and has not been copied from any source without acknowledgement.
- This project has not been submitted for any degree or academic title elsewhere.
- All references and external APIs used have been duly cited and acknowledged.
- The conclusions drawn are based entirely on my own understanding and research.

Place: Bengaluru
Date: July 2026

Signature of the Student
NIKITA
Roll No: 24BCA096 | 4th Semester, BCA

ACKNOWLEDGEMENT

I thank all those who encouraged and supported me throughout the development of this project.

I am grateful to Open-Meteo (open-meteo.com) for providing a free, reliable weather forecasting and geocoding API that forms the data backbone of Nimbus. Without this open API, building a fast and accurate weather dashboard would not have been possible.

I am also grateful to the open-source developer community and educational resources available through MDN Web Docs, Stack Overflow, and YouTube, which guided my understanding of REST API integration, asynchronous JavaScript, the Geolocation API, localStorage management, responsive CSS design, and glassmorphism UI techniques.

I thank my family for their constant support and encouragement throughout the vacation period.

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ABSTRACT

Weather information is something people check multiple times a day, yet many weather websites are cluttered with advertisements, slow to load, and provide an overwhelming amount of data without a clear visual hierarchy. Students and everyday users need a clean, fast, and visually appealing weather dashboard that delivers current conditions, hourly trends, and a multi-day outlook in a single glance.

Nimbus is a responsive, single-page weather dashboard that connects to the Open-Meteo API to provide live geocoding and forecast data for any city in the world. Users can search for a city by name, use their current location via browser geolocation, view a glassmorphism-styled current weather hero card, scroll through an hourly forecast for the next 24 hours, browse a 7-day forecast grid, and review detailed weather highlights including feels-like temperature, humidity, wind speed, sunrise, sunset, and UV index — all within a polished light/dark themed interface.

Key Features:

- City search powered by Open-Meteo's geocoding API with spelling-tolerant fallback suggestions.
- Recent search history saved locally for quick re-access to previously searched cities.
- Location detection using the browser's Geolocation API for one-tap local weather.
- Current weather hero card showing city, date, temperature, condition, and a weather emoji icon.
- Next 24 Hours horizontal scroll strip with hourly temperature and condition icons.
- 7-Day Forecast grid displaying daily highs, lows, and condition icons.
- Weather Highlights panel covering feels-like temperature, humidity, wind speed, sunrise, sunset, and UV index.
- Light/dark theme toggle with a smooth glassmorphism aesthetic.
- Fully responsive, mobile-first layout with smooth scroll-based section transitions.
- Built-in error handling and loading states for a polished UX.

Technologies Used:

HTML5, CSS3, Vanilla JavaScript, Open-Meteo REST API (Geocoding & Forecast), Browser Geolocation API, localStorage API. Deployed on Netlify.

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Chapter 1

INTRODUCTION

1.1 Background

Weather conditions affect almost every decision people make during the day — from choosing what to wear, to planning travel, to deciding whether outdoor activities are advisable. While weather information is freely available, most weather websites and apps are cluttered with advertisements, autoplaying videos, and dense layouts that bury the essential numbers a user actually needs.

Open weather APIs such as Open-Meteo provide free, structured, and license-friendly access to global weather forecast and geocoding data without requiring an API key. A well-designed front-end application built on top of such an API can deliver a faster, cleaner, and more focused weather experience than most commercial weather portals.

1.2 About the Project

Nimbus is a responsive, single-page weather dashboard that uses the Open-Meteo API as its data source to provide city search, current location detection, hourly and 7-day forecasts, and detailed weather highlights, all wrapped in a modern glassmorphism interface with light and dark themes.

The application is built entirely with HTML5, CSS3, and vanilla JavaScript, with all logic embedded directly in a single HTML file. It is deployed as a static site on Netlify and requires no server-side infrastructure.

Live at: https://weather-nimbus-nikita.netlify.app/ — deployed on Netlify.

1.3 Application Architecture

Nimbus follows a single-page application architecture where all HTML structure, CSS styling, and JavaScript logic are contained within one file. The JavaScript layer is organised into functional groups:

- **Geocoding Module**: Resolves a user-entered city name to latitude/longitude coordinates via Open-Meteo's geocoding endpoint, with fallback handling for ambiguous or misspelled city names.
- **Forecast Module**: Fetches current conditions, hourly data for the next 24 hours, and a 7-day daily forecast from Open-Meteo's forecast endpoint.
- **Location Module**: Uses the browser's Geolocation API to detect the user's coordinates and automatically loads local weather.
- **Recent Searches Manager**: Stores and retrieves a list of recently searched cities in localStorage for quick lookup.
- **Theme Manager**: Handles light/dark mode toggling and persists the user's preference in localStorage.

- **UI Renderer**: Updates the hero card, hourly scroll strip, 7-day grid, and highlights panel based on fetched data, including weather condition emoji mapping.

1.4 Scope

- Weather discovery via Open-Meteo API: city search with geocoding and current-location detection.
- Current weather hero card with temperature, condition text, and emoji icon.
- Hourly forecast strip for the next 24 hours.
- 7-day forecast grid with daily highs and lows.
- Weather highlights panel: feels-like, humidity, wind, sunrise, sunset, UV index.
- Recent search history and light/dark theme toggle.
- Fully responsive, mobile-first layout with scroll-friendly sections.

Chapter 2

OBJECTIVES OF THE PROJECT

The following objectives guided the design and development of Nimbus:

1. To build a real-world API-integrated weather dashboard: Integrate Open-Meteo's geocoding and forecast REST API endpoints to provide live current weather, hourly forecasts, and a 7-day outlook for any searchable city in the world.

2. To implement reliable city search with fallback handling: Support city-name search with spelling tolerance and graceful handling of ambiguous or unrecognised city names, guiding the user toward a nearby city, country, or landmark.

3. To deliver a complete current-conditions experience: Present a hero card showing city name, date, current temperature, weather condition, and a representative emoji icon for quick at-a-glance understanding.

4. To provide short-term and extended forecasting: Display an hourly forecast for the next 24 hours in a horizontally scrollable strip, and a 7-day forecast grid showing daily high/low temperatures and conditions.

5. To surface meaningful weather highlights: Build a dedicated highlights panel covering feels-like temperature, humidity percentage, wind speed, sunrise time, sunset time, and UV index.

6. To enable personalised and convenient usage: Allow users to detect their current location via the Geolocation API, and maintain a recent-search history in localStorage for quick re-access to previously viewed cities.

7. To deliver a polished, responsive single-page UI: Design a mobile-first, glassmorphism-styled interface with a light/dark theme toggle, smooth scroll-based section transitions, and clear loading and error states.

Chapter 3

EXISTING SYSTEM

3.1 Current Weather Information Tools

Users currently rely on a variety of platforms to check weather conditions, each with notable limitations:

a) Commercial Weather Websites (AccuWeather, The Weather Channel, Weather.com)

These sites provide detailed forecasts but are heavily monetised with advertisements, autoplaying videos, and pop-up notification prompts. Page load times are slow, and the interface often buries the current temperature beneath promotional banners.

b) Default Phone Weather Apps

Built-in weather apps are convenient but offer limited customisation, often require location permissions for all features, and do not provide a comparable web-based experience that can be accessed instantly from any device or browser.

c) Search Engine Weather Widgets

Searching “weather + city name” on a search engine shows a basic widget, but this provides only minimal information, lacks an hourly breakdown beyond a short window, and offers no persistent history or theme customisation.

d) Third-Party Weather Apps (Weather.com app, AccuWeather app)

Dedicated mobile weather apps often require installation, account sign-in for extended features, and may include intrusive notifications or advertisements, making them less appealing for users who simply want a quick web-based check.

3.2 Limitations Summary

- Commercial weather sites are slow, cluttered, and advertisement-heavy.
- No free, clean, browser-based tool combines city search, location detection, hourly and weekly forecasts, and a modern theme toggle in one lightweight page.
- Recent search history is rarely offered on free weather websites.
- Spelling-tolerant city search with helpful fallback suggestions is uncommon.
- Most platforms require account creation or app installation and are not privacy-respecting.

Chapter 4

PROPOSED SYSTEM

4.1 Overview

Nimbus proposes a clean, fast, single-page weather dashboard that integrates Open-Meteo's open geocoding and forecast APIs for live weather data, and provides a complete personal weather-checking experience — all within a polished, responsive, glassmorphism-styled web application supporting both light and dark themes.

4.2 Advantages

- **Clean, Ad-Free Interface**: No advertisements, no autoplay videos, and no pop-up prompts. Weather content is presented immediately and clearly.
- **Reliable City Search**: Spelling-tolerant geocoding with fallback suggestions for nearby cities, countries, or landmarks when an exact match is not found.
- **Instant Location Detection**: One-tap access to local weather using the browser's Geolocation API, with no manual coordinate entry required.
- **Comprehensive Forecast Views**: Current conditions, next-24-hours hourly strip, and a full 7-day forecast grid in one continuous scroll.
- **Detailed Highlights Panel**: Feels-like temperature, humidity, wind speed, sunrise, sunset, and UV index presented in dedicated highlight cards.
- **No Account Required**: Recent searches and theme preference are persisted in localStorage with no login needed.
- **Responsive and Mobile-Friendly**: Designed mobile-first, with scroll-friendly sections, smooth transitions, and touch-optimised controls.

4.3 API Integration Architecture

Nimbus uses Open-Meteo's free APIs via centralised fetch wrappers:

Endpoint / Pattern	Used For
geocoding-api.open-meteo.com/v1/search?name=	Resolve city name to latitude/longitude with fallback candidates
api.open-meteo.com/v1/forecast (current_weather=true)	Current temperature, condition code, and weather icon
api.open-meteo.com/v1/forecast (hourly=...)	Next 24 hours temperature and condition data
api.open-meteo.com/v1/forecast (daily=...)	7-day forecast: daily high/low and condition
daily=sunrise,sunset,uv_index_max	Sunrise, sunset, and UV index for highlights panel

Browser Geolocation API	Detect user's current coordinates for local weather

Chapter 5

MODULES AND FEATURES

Nimbus is a single-page application comprising several functional sections that flow vertically on one page.

5.1 Header and Search Bar

The top navigation bar displays the Nimbus logo with a cloud icon, a central search input with placeholder text “Search city...”, a settings icon, and a theme toggle (light/dark mode) icon in the top-right corner.

On submitting a search, the app calls the geocoding API to resolve the entered city name to coordinates. If the city is not found, an error toast such as “City not found. Try another name.” is shown, and the app suggests trying a nearby city, country, or landmark.

5.2 Current Weather Hero Card

The hero section is the primary focus of the dashboard, displaying:

- City name and country (e.g., “Chennai, IN” or “Bengaluru, IN”).
- Current date (e.g., “Friday, June 12”).
- A large temperature display in degrees Celsius.
- A short condition description such as “Overcast” or “Heavy Drizzle”.
- A weather condition emoji/icon (cloud, rain, sun, etc.) corresponding to the Open-Meteo weather code.

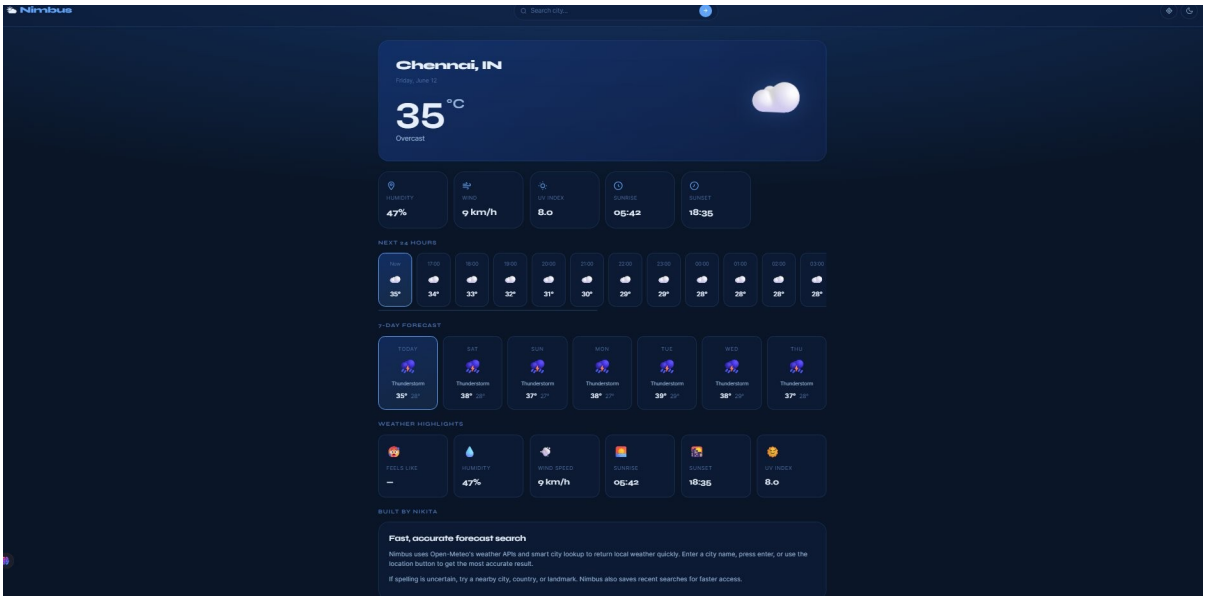


Figure 5.1 – Desktop View: Current Weather Hero Card for Chennai with Quick Stats, Hourly and 7-Day Forecast

5.3 Quick Stats Row

Directly below the hero card, a row of five compact stat cards displays Humidity (%), Wind speed (km/h), UV Index, Sunrise time, and Sunset time, each with its own icon for quick visual scanning.

5.4 Next 24 Hours

A horizontally scrollable strip labelled “NEXT 24 HOURS” shows one card per hour, each containing the hour label (e.g., 17:00, 18:00), a weather condition icon, and the forecast temperature for that hour. The current hour card (“Now”) is visually highlighted with a border to distinguish it from the rest of the strip.

5.5 7-Day Forecast

A grid labelled “7-DAY FORECAST” presents one card per day from today through the following week. Each card shows the day label (TODAY, SAT, SUN, MON, TUE, WED, THU), a weather condition icon (e.g., a thunderstorm icon), and the day's high and low temperatures. The current day's card is highlighted to mark it as “TODAY”.

5.6 Weather Highlights Panel

The “WEATHER HIGHLIGHTS” section presents six detail cards in a responsive grid:

- **Feels Like**: Perceived temperature accounting for wind and humidity.
- **Humidity**: Current relative humidity percentage.
- **Wind Speed**: Current wind speed in km/h.
- **Sunrise**: Time of sunrise for the searched location.
- **Sunset**: Time of sunset for the searched location.
- **UV Index**: Current UV index value, useful for sun-exposure planning.

5.7 About Section — Built by Nikita

A footer-style information panel titled “BUILT BY NIKITA” introduces the app with the heading “Fast, accurate forecast search”, explaining that Nimbus uses Open-Meteo's weather APIs and smart city lookup to return local weather quickly, and that users can enter a city name, press enter, or use the location button for the most accurate result. It also notes that if spelling is uncertain, users should try a nearby city, country, or landmark, and that Nimbus saves recent searches for faster access.

5.8 Mobile Responsive Layout

On smaller screens, Nimbus reflows into a single-column, mobile-first layout. The search bar moves to the top with the app logo, the hero card becomes a full-width glassmorphism panel, and the quick stats collapse into a two-column grid below the hero card.

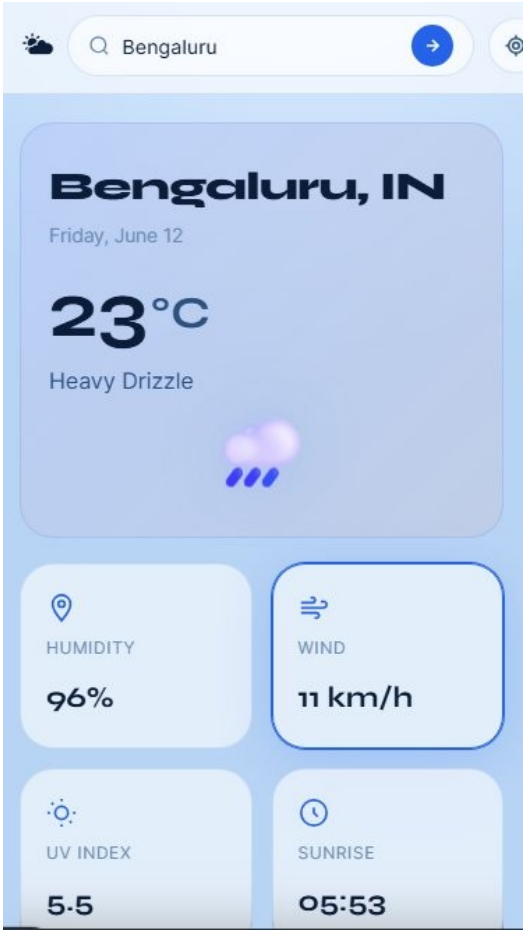


Figure 5.2 – Mobile View: Bengaluru Weather with Heavy Drizzle, Humidity, Wind, UV Index, and Sunrise Cards

5.9 Recent Searches

Every successful city search is stored in localStorage as part of a recent-search list. This allows users to quickly revisit previously searched cities without retyping the name, improving repeat-usage convenience.

5.10 Location Detection

A location icon next to the search bar invokes the browser's Geolocation API. On permission grant, the app retrieves the user's coordinates and performs a reverse geocoding lookup to determine the nearest city, then loads its weather data automatically.

5.11 Light/Dark Theme Toggle

A toggle in the header switches between a light theme (soft blue-grey glassmorphism background with dark text, as seen on mobile) and a dark theme (deep navy background with light text and orange/blue accent highlights, as seen on desktop). The selected theme is saved to localStorage and persists across page reloads.

Chapter 6

TECHNOLOGIES USED

Nimbus was developed using native front-end web technologies and one external API. No build tools, frameworks, or back-end server are used.

6.1 HTML5

HTML5 provides the structural foundation for the single-page Nimbus dashboard. Semantic elements organise the header, hero section, hourly scroll strip, 7-day grid, and highlights panel. Data attributes on forecast cards store metadata used by JavaScript for rendering and updates.

6.2 CSS3

CSS3 delivers Nimbus's glassmorphism interface with translucent panels, blurred backgrounds, and soft shadows. CSS Custom Properties manage both the light and dark theme colour systems, switching via a class toggle on the root element and persisted to localStorage. CSS Grid lays out the 7-day forecast and weather highlights cards. Flexbox handles the header, hero section, and hourly scroll strip. CSS transitions power smooth scroll-based section reveals. The application is fully responsive via CSS media queries across mobile, tablet, and desktop breakpoints.

6.3 JavaScript (ES6+, Vanilla)

All application logic is implemented in Vanilla JavaScript embedded within the single HTML file. The core logic includes: an async geocoding function that resolves city names to coordinates with fallback candidate handling, a forecast-fetching function that retrieves current, hourly, and daily weather data in one combined request, a weather-code-to-emoji mapping function for visual condition icons, a recent-searches manager backed by localStorage, a geolocation handler for current-location weather, a theme manager for light/dark mode persistence, and render functions that populate the hero card, hourly strip, 7-day grid, and highlights panel.

6.4 Open-Meteo REST API

Open-Meteo (open-meteo.com) is a free, open-source weather API providing global geocoding and forecast data without requiring an API key. Nimbus uses the geocoding endpoint to convert city names into coordinates, and the forecast endpoint with combined current_weather, hourly, and daily parameters to retrieve all data needed for the dashboard in minimal requests.

6.5 Browser Geolocation API

The Geolocation API allows Nimbus to request the user's current coordinates directly from the browser, enabling one-tap access to local weather without manual city entry.

6.6 localStorage API

The Web Storage API's localStorage interface persists user-specific state across sessions, storing the recent-search list and the user's light/dark theme preference.

6.7 Netlify

Nimbus is deployed as a static single-page site on Netlify, providing HTTPS, a custom subdomain, and global CDN distribution. As a static site with no server-side code, Netlify's free tier is fully sufficient.

1	HTML5	Markup	Single-page structure, semantic sections, data attributes
2	CSS3	Styling	Glassmorphism UI, light/dark themes, responsive grid
3	JavaScript ES6+	Programming	API calls, geolocation, localStorage, rendering
4	Open-Meteo API	External REST API	Geocoding and forecast data source
5	Geolocation API	Browser API	Detect current location for local weather
6	localStorage	Browser API	Recent searches, theme preference
7	Netlify	Platform	Static single-page hosting and CDN

Chapter 7

CONCLUSION

Nimbus is a well-architected weather dashboard that successfully integrates a live REST API with thoughtful UI design to deliver a complete, glanceable weather companion for everyday use. By combining Open-Meteo's geocoding and forecast data with a glassmorphism-styled interface, recent-search history, and location-based detection, the project delivers a genuinely useful tool that addresses real limitations of cluttered commercial weather websites.

The technical breadth of Nimbus is notable for a single-file project. Integrating geocoding and multi-parameter forecast endpoints, building a weather-code-to-icon mapping system, implementing the Geolocation API for one-tap local weather, managing recent-search history and theme preference through localStorage, and delivering a fully responsive light/dark themed UI — all in Vanilla JavaScript without any framework or build step — represents a solid front-end development achievement.

The decision to combine current conditions, an hourly scroll strip, a 7-day grid, and a detailed highlights panel into one continuously scrollable page demonstrates thoughtful information architecture that prioritises the information users care about most, presented in the order they are likely to need it.

On a personal level, building Nimbus during the vacation period has been a valuable experience in working with real-world external APIs, designing a clean single-page application architecture, and creating a polished, deployable product with a distinctive visual style.

In conclusion, Nimbus is a polished, feature-rich, and publicly deployed weather dashboard that fully achieves its stated objectives and stands as a strong demonstration of API integration skills, front-end design sensibility, and independent learning initiative.

Live Project: https://weather-nimbus-nikita.netlify.app/